SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I AB. SAI SARANYA (192126O020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sai Saranya.ab

Criteria	Weightage (%)	Q1 6/10 marks	Q2 6/10 marks	Q3 6/10 marks	Q4 6/10 marks	Q5
Understanding of Scenario	20%	2	2	1	2	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	1	2	1	2
Decision Making	20%	1	2	2	1	1
Clarity of Response	15%	1	1	1	2	1

12/15

6

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification	20%	Accurately	Identifies	Identifies	Unable to

of Key Issues		identifies all relevant issues and factors involved	most key issues with minor omissions	limited issues; some irrelevant points	identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS TO SCENARIO-BASED QUESTIONS

Course: Computer Networks (CSA07) | CO2 | Assessment Tool 2

Name: **AB Sai Saranya** | Register Number: **1921260020**

Question 1: Smart Campus subnet allocation

Scenario: The Smart Campus has the block 192.168.100.0/24. School of Computing gets a /25, School of Electronics a /26 and School of Mechanical Engineering a /27. Find the network and broadcast address, host range and usable hosts of each subnet, and the unused range.

Understanding the problem

The /24 block has 8 host bits, so 256 addresses (192.168.100.0 to 192.168.100.255). The three masks are fixed, so I only have to place the blocks one after another without overlap. I start with the largest block (/25) so that every block begins on a proper boundary, which is a multiple of its own size. In each subnet the first address is the network address, the last is the broadcast address, and the addresses in between are usable hosts ($2^n - 2$).

Allocation

Department	Mask	Network address	Valid host range	Broadcast address	Usable hosts
Computing /25	255.255.255.128	192.168.100.0	192.168.100.1 – .126	192.168.100.127	$2^7-2 = 126$
Electronics /26	255.255.255.192	192.168.100.128	192.168.100.129 – .190	192.168.100.191	$2^6-2 = 62$
Mechanical /27	255.255.255.224	192.168.100.192	192.168.100.193 – .222	192.168.100.223	$2^5-2 = 30$

Unused address range

The last address used is 192.168.100.223, so the free range is **192.168.100.224 – 192.168.100.255**. That is 32 addresses (128 + 64 + 32 = 224 used out of 256), which is exactly one /27 block. The campus can use it later, for example for a new school or for the server and Wi-Fi management network.

Conclusion

The three subnets fit neatly one after another without overlapping, and Computing, which probably has the most devices, gets the biggest subnet (126 hosts). Since a whole /27 is still free, the design leaves space for future growth.

Question 2: Software company headquarters (10.10.0.0/16)

Scenario: The company has 10.10.0.0/16. Software Development uses /24, Quality Assurance /25, Human Resources /26 and Executive Management /27. Find the network and broadcast addresses, valid hosts, usable hosts and the unused range.

Understanding the problem

A /16 has 16 host bits, which means 65,536 addresses (10.10.0.0 – 10.10.255.255). The departments need different sizes, so this is a VLSM design. I allocate from the largest to the smallest block so that no space is lost and each network address is a multiple of its block size.

Allocation

Department	Network address	Valid host range	Broadcast address	Usable hosts
Software Development /24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance /25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources /26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management /27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Reasoning

- Development takes the whole of 10.10.0.0/24 (256 addresses). The next free address is 10.10.1.0.
- QA takes 10.10.1.0 – .127 (128 addresses). HR then starts at .128 (a multiple of 64) and ends at .191.
- Executive Management starts at .192 (a multiple of 32) and ends at .223.

Unused address range

Total used = 256 + 128 + 64 + 32 = 480 addresses, ending at 10.10.1.223. The unused range is therefore **10.10.1.224 – 10.10.255.255**. This is 32 addresses (10.10.1.224 – .255) plus 254 × 256 = 65,024 addresses (10.10.2.0 – 10.10.255.255), giving 65,056 in total. As a check, 65,536 – 480 = 65,056.

Conclusion

All four departments fit inside 10.10.0.0 – 10.10.1.223, which is only a small part of the /16. Because the blocks are contiguous and aligned, the whole group can later be summarised as one route (10.10.0.0/23 covers 10.10.0.0 – 10.10.1.255), and the large unused space leaves plenty of room for new departments or branch offices.

Question 3: Smart hospital IPv4 to IPv6 migration

Scenario: The hospital is given the IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000/64. Compress the address, expand it again, identify the network prefix, and calculate the number of host addresses.

Understanding the problem

An IPv6 address has 128 bits, written as eight 16-bit groups in hexadecimal. A /64 prefix means the first 64 bits (four groups) identify the network and the remaining 64 bits are for devices. The hospital has thousands of medical devices and IoT sensors, so I also want to see whether the subnet is large enough.

1. Compressed notation

- Remove leading zeros in every group: 2001:db8:abcd:0:0:0:0:0
- Replace the single longest run of all-zero groups (the last five groups) with "::": **2001:db8:abcd::/64**

2. Expanded (full) form

2001:db8:abcd:: has only three groups written, so "::" stands for 8 – 3 = 5 zero groups. Putting them back and padding each group to four digits gives **2001:0db8:abcd:0000:0000:0000:0000:0000**, which is the original address, so the compression is correct.

3. Network prefix

Part	Groups	Value
Network prefix (64 bits)	1 – 4	2001:0db8:abcd:0000
Interface ID (64 bits)	5 – 8	0000:0000:0000:0000

The network prefix is **2001:db8:abcd:0::/64**, normally written **2001:db8:abcd::/64**. The given address has an all-zero interface ID, so it is the network address itself, not a device address.

4. Total host addresses

Host bits = 128 – 64 = 64, so the subnet holds 2^{64} = **18,446,744,073,709,551,616** addresses (about 1.8×10^{19}). IPv6 has no broadcast address, so nothing is subtracted for it. In practice the all-zero interface ID is reserved as the Subnet-Router anycast address, so it should not be given to a normal device. The two last points of the question both ask for the number of host addresses in this network, so one answer covers both.

Conclusion

One /64 is far more than the hospital needs, even with thousands of sensors, so the whole hospital could use a single subnet. A better design is to give each ward or device type (for example patient monitors, imaging and IoT sensors) its own /64 from the hospital's larger allocation, which keeps traffic separated and security rules simple. A /64 is also the size required for SLAAC auto-configuration.

Question 4: ISP core router and packet forwarding

Scenario: A core router has the entries 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. A packet arrives for 192.168.2.45. Find the destination subnet, the matching entry, the next hop / interface and the default route.

Understanding the problem

A router forwards a packet by comparing the destination address with the prefixes in its routing table and choosing the **longest prefix match**, that is, the matching entry with the most network bits. Here all three entries are /24, so the first 24 bits (the first three octets) decide the match.

1. Destination subnet

The destination 192.168.2.45 with the mask 255.255.255.0 gives network 192.168.2.0 (45 is in the host part). The packet therefore belongs to the subnet **192.168.2.0/24**, whose range is 192.168.2.0 – 192.168.2.255 (usable hosts .1 – .254, broadcast .255).

2. Matching routing table entry

Routing entry	First three octets	Matches 192.168.2.45?
192.168.1.0/24	192.168.1	No (1 ≠ 2)
192.168.2.0/24	192.168.2	**Yes**
192.168.3.0/24	192.168.3	No (3 ≠ 2)

Only **192.168.2.0/24** matches, so this is the entry selected.

3. Next hop / outgoing interface

The question lists only the destination prefixes, so the actual next-hop address and interface name are not given and I have not invented them. The packet is sent out of the interface (or to the next-hop router) that the 192.168.2.0/24 entry points to, which is the link towards that customer's network. If the customer network is directly connected, the router ARPs for 192.168.2.45 and delivers the frame straight to the host; if the customer is behind another router, the packet goes to that router's address. For example, the table could read: 192.168.2.0/24 via interface Gi0/1 (an assumed name used only for illustration).

4. Default route

If the destination matches none of the three entries (for example 8.8.8.8 or 192.168.4.10), the router uses the default route **0.0.0.0/0**. It matches every address with zero network bits, so it is always the last choice. On an ISP core router it would point to the upstream provider or Internet gateway, and the router configuration would be something like *ip route 0.0.0.0 0.0.0.0 <next-hop address>*. Without it, such packets would be dropped and the sender would get a "destination unreachable" message.

Conclusion

Packet 192.168.2.45 belongs to 192.168.2.0/24, matches only the second entry, and is forwarded through that entry's next hop. Anything unknown goes through 0.0.0.0/0.

Question 5: Administration and Finance LANs connected by a router

Scenario: Administration LAN is 192.168.10.0/26 and Finance LAN is 192.168.10.64/26, joined by a Layer-3 router. The router interfaces are 192.168.10.1 (Administration) and 192.168.10.65 (Finance). Find the network and broadcast addresses, the host ranges, suggest three computer addresses in each LAN, and name the default gateways.

Understanding the problem

A /26 mask is 255.255.255.192, so each LAN has 6 host bits and a block size of 64. The two LANs are two separate subnets inside the same /24: 192.168.10.0 – .63 and 192.168.10.64 – .127. Each has $2^6 - 2 = 62$ usable hosts.

Network details

	Administration LAN	Finance LAN
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Subnet mask	255.255.255.192	255.255.255.192
Router interface (gateway)	192.168.10.1	192.168.10.65

Suggested computer addresses

The first address of each range is already used by the router interface, so the computers start from the second address. All hosts use mask 255.255.255.192.

Computer	Administration LAN	Finance LAN
PC1	192.168.10.2	192.168.10.66
PC2	192.168.10.3	192.168.10.67
PC3	192.168.10.4	192.168.10.68

Default gateway

- Hosts in the Administration LAN use **192.168.10.1** as the default gateway.
- Hosts in the Finance LAN use **192.168.10.65** as the default gateway.

The gateway must always be an address in the host's own subnet, because a PC can reach it directly at Layer 2. When a PC in one LAN, such as 192.168.10.2, wants to talk to a PC in the other LAN, such as 192.168.10.66, it sees that the destination is outside its own network (using the mask), so it sends the packet to its gateway. The router then forwards it out of the other interface to the Finance LAN. Traffic between two PCs in the same LAN stays inside that LAN and does not need the router.

Conclusion

The two /26 subnets do not overlap, the router interfaces (.1 and .65) are valid host addresses of their own subnets, and each LAN has 62 usable addresses, with 58 still free in each after using three PCs and the gateway. The suggested addresses work correctly with the given gateways. Remember that .0, .63, .64 and .127 are network or broadcast addresses and must not be given to computers.